

NDN Play

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Motivation

- Getting started with NDN is not easy
- Visual demonstrations are nice
- Need tools to showcase NDN to newcomers
 - Students
 - People without a technical background

Introducing NDN Play

- Playground for NDN
- Runs completely in the browser
 - Deployed on GitHub pages
- Simulates an NDN network
- Pretty visualization
- Open source - <https://github.com/pulsejet/ndn-play>
 - Examples available

<https://play.ndn.today/>

Features of NDN Play

- Visual topology editor
 - Can export/import MiniNDN configuration
- Simulation of NDN network
- Run “real” NDNs code on simulated nodes
 - Can run a simple NDN experiments in the browser
- Capture network traffic
- Visualize network traffic
 - Identify hotspots
 - TLV visualization
- Dump and share experiment results

Ongoing and Future Work

- Integration with MiniNDN
 - Live packet capture
 - Live topology manipulation
 - Status reports
 - Network utilization graphs / reports
- Security and trust visualization

Demo

<https://play.ndn.today>

NDN Play GUI

NDN-Play
Global Operations

Compute Routes
Compute Trust

Default Latency (ms)
10

Default Loss (%)
0

Content Store Size
500

Latency Slowdown Multiplier
10

☐ Enable Packet Capture (all nodes)

Experiment Dump:
Load
Generate

MiniNDN Config:
Load

Topology Trust Schema Trust Visualizer

Edit

```
graph TD; D1((D)) --- D2((D)); D2 --- A((A)); A --- E((E)); E --- D3((D)); D3 --- M((M)); M --- B((B)); M --- C((C));
```

Console TLV Visualizer Traffic Replay

Computing trust
Computing routes

Topology Editor

Topology

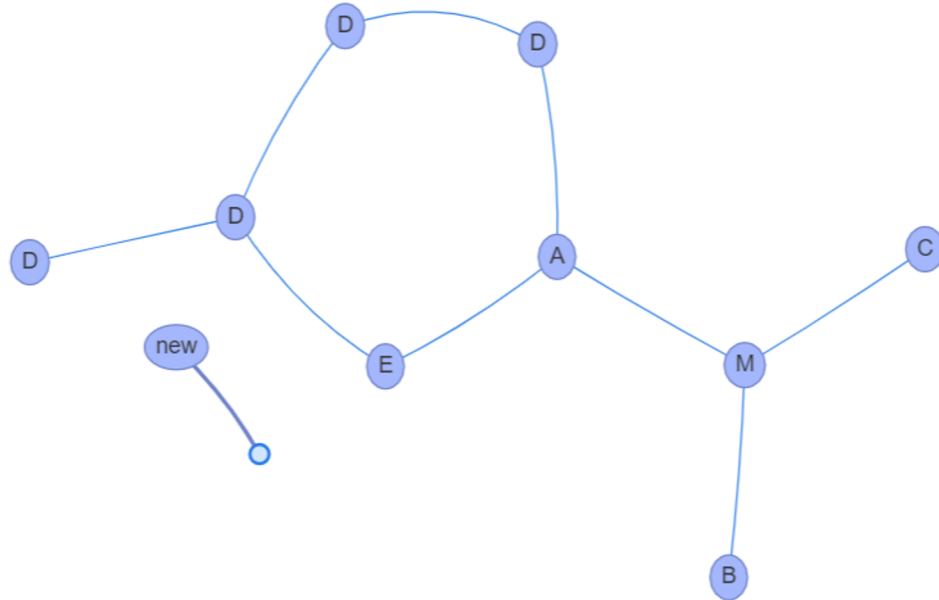
Trust Schema

Trust Visualizer



Back

Click on a node and drag the edge to another node to connect them.



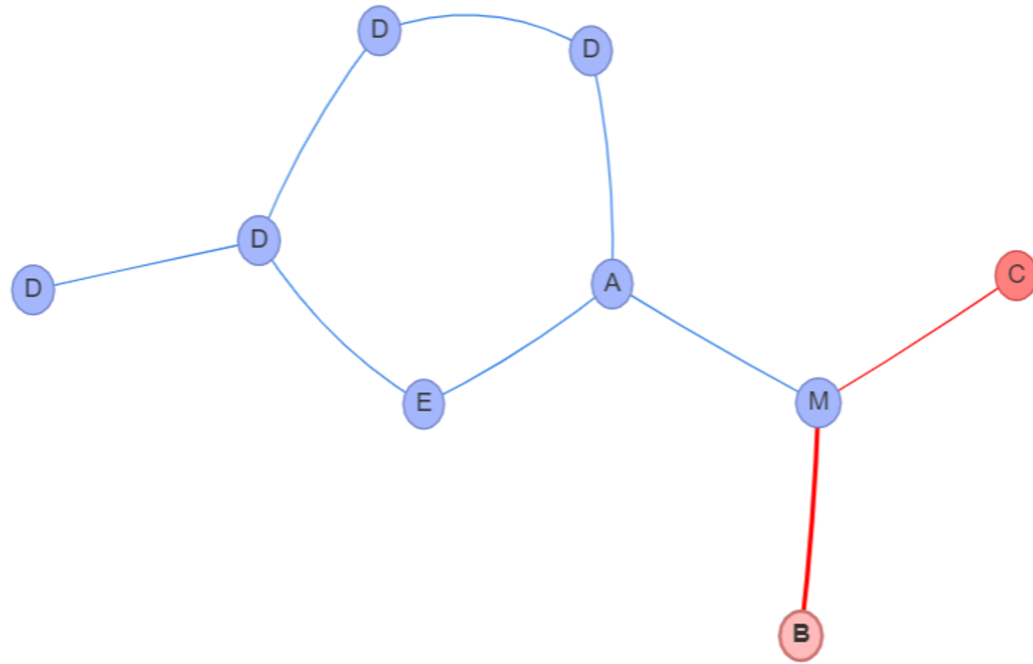
Code Editor with Syntax Completion

Topology Editor Trust Schema Trust Visualizer

```
1 // Producer
2 const { Data, Interest } = ndn.packet;
3 const { fromUtf8, toUtf8 } = ndn.tlv;
4
5 // Create endpoint
6 const endpoint = node.nfw.getEndpoint({ secure: false });
7 var myProducer = endpoint.produce('/ndn/producer/test', async (interest) => {
8   const data = new Data(interest.name, Data.FreshnessPeriod(500));
9   data.content = toUtf8("Hello from NDNTs Producer");
10  return data;
11 });
12
13 endpoint.
14
```

 consume (method) t.EndpointConsumer.consum...
 fw
 opts
 produce

Traffic Visualization



Console	TLV Visualizer	Packet Capture	Traffic Replay
Timestamp	Type	Length	Name
354883 ms	Data [B → M]	847	/ndn/multicast/psync/%00%00%00%00%00%00%00%00%00%
354884 ms	Interest [B → M]	778	/ndn/multicast/psync/%00%00%00%00%00%00%00%00%00%
354886 ms	Data [B → M]	867	/ndn/multicast/psync/%00%00%00%00%00%00%00%00%00%
354887 ms	Data [B → M]	887	/ndn/multicast/psync/%00%00%00%00%00%00%00%00%00%
354888 ms	Interest [B → M]	765	/ndn/multicast/psync/%00%00%00%00%00%00%00%00%00%
355093 ms	Interest [A → M]	778	/ndn/multicast/psync/%00%00%00%00%00%00%00%00%00%
355096 ms	Data [A → M]	867	/ndn/multicast/psync/%00%00%00%00%00%00%00%00%00%

TLV Visualization

[Console](#)
[TLV Visualizer](#)
[Packet Capture](#)
[Traffic Replay](#)

Data 843

Name 764

GenericNameComponent 3 ndn

GenericNameComponent 9 multicast

GenericNameComponent 5 psync

```
GenericNameComponent 720 %00%00%00%00%00%00%00%00%00%00%00 ...
```

GenericNameComponent 0

```
GenericNameComponent 9 %FD%00%00%01%7C%C4%C7h%DA
```

```
GenericNameComponent 2 %00%00
```

MetaInfo	10	FreshnessPeriod	2	0x03e8
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FinalBlockId 4

GenericNameComponent 2 %00%00

Content 22

AppPrivateBlock1 20

Name 18

GenericNameComponent 3 ndn

GenericNameComponent 1 B

```
GenericNameComponent 5 psync
```

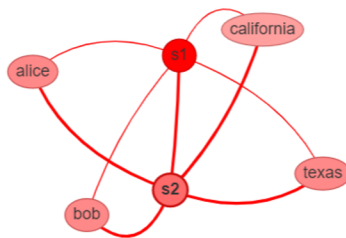
GenericNameComponent 1 %17

SignatureInfo	3	SignatureType	1	0x00
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```
SignatureValue 32 0xba124ee0747cbe2375d59142bda377 ...
```

- ☐ Attempt to decode unknown TLV types

MiniNDN Integration



Console TLV Visualizer <u>Packet Capture</u> Traffic Replay				
Timestamp	Type	Length	Name	
1631225453899 ms	Interest [california → s2]	85	/localnet/mbpsDemo/pub/201=.../2=%ED%22%C3%1E%90%8C%FC2Y%23u%EE%B6%DE%C0%87%E%1 ...	
1631225453899 ms	Interest [california → s2]	659	/localnet/mbpsDemo/cert/201=%CA%2B%07%26%08%05myNet%08%08mbpsDemo%08%07gateway%0 ...	
1631225453899 ms	Interest [california → s2]	213	/localnet/mbpsDemo/key/201=.../2=%22%B8%C4%C3%25%29%B1%A0a%EC8%BB%DB%2B%E%3EA ...	
1631225453919 ms	Interest [s2 → s1]	85	/localnet/mbpsDemo/pub/201=.../2=%ED%22%C3%1E%90%8C%FC2Y%23u%EE%B6%DE%C0%87%E%1 ...	
1631225453919 ms	Interest [s2 → s1]	659	/localnet/mbpsDemo/cert/201=%CA%2B%07%26%08%05myNet%08%08mbpsDemo%08%07gateway%0 ...	
1631225453919 ms	Interest [s2 → s1]	213	/localnet/mbpsDemo/key/201=.../2=%22%B8%C4%C3%25%29%B1%A0a%EC8%BB%DB%2B%E%3EA ...	
1631225453919 ms	Interest [s1 → s2]	85	/localnet/mbpsDemo/pub/201=.../2=%ED%22%C3%1E%90%8C%FC2Y%23u%EE%B6%DE%C0%87%E%1 ...	
1631225453919 ms	Interest [s1 → s2]	659	/localnet/mbpsDemo/cert/201=%CA%2B%07%26%08%05myNet%08%08mbpsDemo%08%07gateway%0 ...	
1631225453919 ms	Interest [s1 → s2]	213	/localnet/mbpsDemo/key/201=.../2=%22%B8%C4%C3%25%29%B1%A0a%EC8%BB%DB%2B%E%3EA ...	
1631225453929 ms	Interest [s2 → texas]	85	/localnet/mbpsDemo/pub/201=.../2=%ED%22%C3%1E%90%8C%FC2Y%23u%EE%B6%DE%C0%87%E%1 ...	
1631225453929 ms	Interest [s2 → texas]	659	/localnet/mbpsDemo/cert/201=%CA%2B%07%26%08%05myNet%08%08mbpsDemo%08%07gateway%0 ...	
1631225453929 ms	Interest [s2 → texas]	213	/localnet/mbpsDemo/key/201=.../2=%22%B8%C4%C3%25%29%B1%A0a%EC8%BB%DB%2B%E%3EA ...	

Thank you!
Q&A